

e& Business Network Operations Guide

Network Security Best Practices and Procedures

1. Introduction

Network security is a critical component of e& Business Network Operations (BNO). This document outlines security policies, procedures, and best practices for protecting the telecommunications infrastructure from threats and vulnerabilities. All BNO personnel must adhere to these guidelines to ensure the integrity, confidentiality, and availability of network services.

2. Security Architecture

The e& network security architecture implements defense-in-depth principles with multiple layers of protection. The security perimeter includes firewalls, intrusion detection systems (IDS), intrusion prevention systems (IPS), and distributed denial-of-service (DDoS) mitigation systems. All security devices are configured in high-availability mode with automatic failover capabilities.

2.1 Perimeter Security

Perimeter firewalls are deployed at all network entry points including internet gateways, data center interconnects, and customer access points. Firewall rules follow the principle of least privilege, allowing only necessary traffic. All firewall rules are reviewed quarterly and documented with business justification. Default deny policies are enforced on all firewall interfaces.

2.2 Internal Network Segmentation

Internal networks are segmented using VLANs and access control lists (ACLs) to limit lateral movement in case of a security breach. Critical systems are isolated in separate network segments with restricted access. Inter-segment communication is controlled through firewall rules and monitored for anomalies.

3. Access Control and Authentication

Access to network devices and management systems is strictly controlled. All access must be authenticated and authorized. Multi-factor authentication (MFA) is required for all administrative access to critical systems. User accounts are provisioned with minimum necessary privileges following the principle of least privilege.

3.1 Device Access Management

Network device access is managed through TACACS+ authentication servers. All user accounts are created with role-based access control (RBAC). Privileged access requires additional approval and is granted on a time-limited basis. All access attempts are logged and monitored for suspicious activity. Failed authentication attempts trigger alerts after three consecutive failures.

3.2 Password Policies

Strong password policies are enforced across all systems: - Minimum length: 12 characters - Must include uppercase, lowercase, numbers, and special characters - Password expiration: 90 days - Password history: Cannot reuse last 5 passwords - Account lockout: 5 failed attempts lock account for 30 minutes

4. Vulnerability Management

Regular vulnerability assessments are conducted to identify security weaknesses in network infrastructure. Critical and high-severity vulnerabilities are patched within 7 days of identification. Medium-severity vulnerabilities are addressed within 30 days. All patches are tested in lab environments before deployment to production.

4.1 Security Patch Management

Security patches are released monthly during standard maintenance windows. Emergency patches for critical vulnerabilities may be deployed outside maintenance windows with proper approval. All patches are tracked in the change management system with rollback procedures documented. Patch deployment is verified through automated testing and manual verification.

4.2 Security Scanning

Automated security scanning is performed weekly on all network devices. Scans check for known vulnerabilities, misconfigurations, and compliance violations. Scan results are reviewed by the security team and remediation plans are developed for identified issues. Critical findings are escalated immediately to the network operations manager.

5. Incident Response and Forensics

Security incidents are handled according to the BNO incident response plan. All security incidents are logged, investigated, and documented. Critical security incidents are escalated immediately to the security operations center (SOC) and network operations manager. Forensic analysis is performed for all major security incidents to determine root cause and prevent recurrence.

5.1 Security Incident Classification

Security incidents are classified based on severity:

Severity	Description	Response Time
Critical	Active breach, data exfiltration, or service disruption	15 minutes
High	Unauthorized access detected, malware infection	1 hour
Medium	Suspicious activity, policy violation	4 hours
Low	Security alerts, minor policy violations	1 business day

5.2 Incident Containment

When a security incident is detected, immediate containment actions are taken to prevent further damage. Affected systems may be isolated from the network. Compromised accounts are disabled immediately. Network traffic from affected sources is blocked at firewalls. Containment actions are documented and communicated to stakeholders.

6. Network Monitoring and Threat Detection

Continuous monitoring of network traffic is performed to detect security threats. Security information and event management (SIEM) systems aggregate logs from all network devices and security tools. Automated alerts are configured for known attack patterns and anomalies. Security analysts review alerts and investigate potential threats.

6.1 Threat Intelligence

Threat intelligence feeds are integrated into security monitoring systems to identify known malicious IP addresses, domains, and file hashes. Indicators of compromise (IOCs) are automatically blocked at firewalls and IPS devices. Threat intelligence is updated daily from multiple sources including commercial feeds and open-source intelligence.

6.2 Log Management

All network devices generate security logs that are forwarded to centralized log management systems. Logs are retained for a minimum of 90 days with critical security logs retained for 1 year. Log analysis tools are used to correlate events across multiple systems and identify security incidents. Regular log reviews are performed to identify trends and anomalies.

7. Compliance and Audit

BNO security practices comply with international standards including ISO 27001 and local regulatory requirements. Regular security audits are conducted by internal audit teams and external auditors. Audit findings are tracked and remediated according to established timelines. Security compliance reports are prepared quarterly for management review.

7.1 Security Documentation

All security policies, procedures, and configurations are documented and maintained in the security documentation repository. Documentation is reviewed and updated annually or when significant changes occur. Access to security documentation is restricted to authorized personnel only. Version control is maintained for all security documentation.

8. Security Training and Awareness

All BNO personnel receive security awareness training upon hire and annually thereafter. Training covers security policies, threat awareness, and incident reporting procedures. Specialized security training is provided to personnel with security responsibilities. Security awareness campaigns are conducted quarterly to reinforce security practices.

9. Contact Information

For security-related issues, contact:

- Security Operations Center (SOC): +971-XX-XXX-XXXX
- Email: bno-security@eand.ae
- Emergency Security Hotline: +971-XX-XXX-XXXX (24/7)

- Security Manager: Available during business hours

For security incidents, report immediately through the security incident reporting portal or contact the SOC directly.

Appendix A: Security Checklist

Pre-deployment security checklist for new network devices:

- ☐ Change default passwords
- ☐ Disable unused services and ports
- ☐ Enable logging and forward to SIEM
- ☐ Configure access control (TACACS+)
- ☐ Apply security patches
- ☐ Configure firewall rules
- ☐ Enable encryption for management access
- ☐ Document in asset inventory
- ☐ Add to monitoring systems

Appendix B: Common Security Commands

Useful commands for security troubleshooting:

- Show Access Lists: `show access-lists`
- Show Firewall Rules: `show firewall rules`
- Show Active Sessions: `show users`
- Show Failed Logins: `show log | grep "failed"`
- Show Security Events: `show security events`
- Block IP Address: `firewall block ip [address]`
- Unblock IP Address: `firewall unblock ip [address]`